

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 1

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Topic

Scope of the Standard Theory

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Scope of the Standard Theory. State all assumptions and conventions.

$$g_{ij} = \delta_{ij}$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\nabla^2 \Phi = 4\pi G \rho$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$R_a = (O_a - T_a(\theta))/\sigma_a$$

Show enough intermediate work that another student can reproduce the calculation.